

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.7: Volume and Capacity — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.7: Volume and Capacity
Driving Question	DRIVING QUESTION: Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Kericho Container Puzzle: Anchoring the Phenomenon & Launching the Driving Question	Three containers used by Kericho tea farmers — a cylindrical drum (A), a conical sisal cover (B), and a metal frustum bucket (C) — all with roughly the same height and opening diameter, were shown filled to the brim. Container A held ≈ 9.4 L, Container B held ≈ 3.1 L, and Container C held ≈ 6.5 L. The cylinder held almost exactly three times as much as the cone.	Volume is measured in cm^3 or m^3 ; capacity is measured in mL or L. The conversion relationships are: $1 \text{ cm}^3 = 1 \text{ mL}$, $1\,000 \text{ cm}^3 = 1 \text{ L}$, and $1 \text{ m}^3 = 1\,000 \text{ L}$. The ratio of the cylinder's capacity to the cone's capacity (same height, same base) is $3 : 1$ — a fact that is not yet explained but will be investigated across this unit.	We cannot yet fully explain why containers of nearly the same height and opening diameter hold such different amounts. Our current best explanation is that the cone narrows to a point, leaving less interior space, while the cylinder stays the same width all the way down. We have posted our questions on the Driving Question Board and will build toward a complete mathematical explanation in the lessons ahead.
Lesson 2 Cone vs. Cylinder — Deriving the Cone Volume Formula	Learners filled a cone with water and poured it into a matching cylinder three times, directly observing that the cylinder was exactly full after the third pour — demonstrating $V(\text{cylinder}) = 3 \times V(\text{cone})$ using containers that represent the Kericho tea-farmer drum and conical sisal cover.	Learners derived the cone volume formula $V = \frac{1}{3}\pi r^2 h$ algebraically from the observed 1:3 ratio, calculated the capacity of the Kericho conical cover ($r = 21 \text{ cm}$, $h = 42 \text{ cm}$) as approximately $19\,373 \text{ cm}^3 \approx 19.4$ litres, and confirmed it holds exactly one-third the volume of the matching cylindrical drum.	Learners explained in writing why the conical sisal cover holds far less water than the cylindrical drum despite identical height and opening width, and updated their cumulative container diagram with the formula, the calculated capacity, and a $\times 3$ ratio arrow — leaving a deliberate question mark on the metal bucket (frustum) as the next mathematical puzzle to solve.
Lesson 3 Volume of Pyramids: Unlocking the One-Third	Learners filled a card-stock square-based pyramid with rice and poured it into an identical-base-and-height cuboid, repeating	$V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$; the same one-third relationship that links a cone to its cylinder also links a square-	The one-third factor in both $V(\text{cone}) = \frac{1}{3} \times \pi r^2 \times h$ and $V(\text{pyramid}) = \frac{1}{3} \times b^2 \times h$ reflects a geometric pattern: any pointed solid (apex

Relationship	until the cuboid was full. They recorded that it took 3 full pyramids to fill 1 cuboid, and used a ruler to measure the base and height of their pyramid model.	based pyramid to its cuboid. Learners also practised converting pyramid volumes from cm^3 and m^3 into litres ($1 \text{ m}^3 = 1\,000 \text{ litres}$) using grain-store problems from Eldoret, Kisumu, and the Rift Valley.	tapering to a point) holds exactly one-third of its surrounding box, regardless of whether the base is circular or square. This explains why the pyramidal grain stores used by Kenyan farmers hold far less grain than a similarly sized cuboid store — and raises the question of how to compute the volume of the frustum-shaped metal bucket, which lies between these two extremes.
Lesson 4 Introducing Frustums of Cones: Deriving the Volume Formula and Solving the Bucket Mystery	Learners physically measured a frustum-shaped metal bucket (top radius R , bottom radius r , vertical height h) and observed a cone model placed alongside it, noticing that the frustum is a large cone with a small cone removed from the bottom.	Learners derived $V(\text{frustum of cone}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ by subtracting $V(\text{small cone})$ from $V(\text{large cone})$, using similar-triangle reasoning to eliminate the unknown full height H . They calculated the Kericho bucket's capacity in litres and placed it correctly between the cone ($\sim 6.16 \text{ L}$) and the cylinder ($\sim 18.47 \text{ L}$), resolving the anchoring phenomenon.	The frustum bucket holds more water than the cone because removing the narrow tip of the cone discards very little volume — the tip contributes almost nothing — while retaining the wide bottom section. The three terms R^2 , Rr , r^2 in the formula arise algebraically from expanding the subtraction of two cone volumes and reflect the interplay between the large radius, the small radius, and their geometric mean. This is why containers of the same height and top diameter can hold dramatically different amounts depending on their shape.
Lesson 5 Volume of Frustums of Pyramids — Designing the Perfect Grain-Storage Bin	Learners observed a side-by-side derivation comparing the volume formula for a pyramidal frustum, $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$, with the conical frustum formula from Lesson 4, revealing that both share an identical mathematical structure — one-third times height times the sum of the top area, the bottom area, and their geometric mean. They also observed scale drawings of the Kericho tea-leaf bucket (conical frustum) and a Rift Valley grain-storage bin (pyramidal frustum) with the same height but different base shapes.	Learners determined that the volume of a frustum of a square or rectangular pyramid is given by $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1A_2})$, where A_1 and A_2 are the areas of the two parallel faces and h is the perpendicular height. They computed that a pyramidal grain-storage bin (top side 30 cm, bottom side 20 cm, height 40 cm) has a volume of approximately $25\,333 \text{ cm}^3 \approx 25.33 \text{ litres}$ and can hold approximately 18.24 kg of dry maize (density 0.72 kg/litre). They also applied the volume scale factor $k^3 = 2$ (from Sub-Strand 2.1) to determine that doubling the capacity while keeping the height fixed requires scaling each side length by $\sqrt[3]{2} \approx 1.26$.	Learners explained that the Kericho tea farmers' metal bucket (conical frustum) and the Rift Valley pyramidal grain bin hold very different amounts of water even at the same height because their base areas — and therefore the values of A_1 , A_2 , and $\sqrt{A_1A_2}$ — are fundamentally different: circular bases are governed by πr^2 , while square bases are governed by s^2 . The formula structure is the same, but the base-area calculation changes depending on the shape of the cross-section, which is why containers that look similar in height can hold dramatically different volumes. This advances the driving question: to design a container of exact capacity, a farmer must choose the base shape and dimensions carefully, not just the height.
Lesson 6 Problem Solving with Real Containers: Apply & Q&B Update	Three different containers used by Kericho tea farmers — a cone, a cylinder, and a frustum-shaped metal bucket — appear nearly the same size when placed side by	The volume of a cone is exactly one-third the volume of a cylinder with the same base radius and height ($V = \frac{1}{3}\pi r^2 h$ vs $V = \pi r^2 h$). A frustum of a cone has volume $V =$	The three Kericho containers look similar because they share roughly the same height and opening diameter, but their shapes distribute that space very differently.

	side, yet when filled with water the cylinder holds three times as much as the cone, and the frustum holds an amount between the two.	$\frac{1}{3}\pi h(R^2 + Rr + r^2)$, which is always greater than the cone alone but less than the full cylinder. These formulas allow us to calculate and compare the exact capacity of any container — including sizing a Turkana water tank, a Kisumu githeri pot, or counting how many Kericho tea cups fill a storage drum.	The cylinder fills every cross-section from base to top with a full circle of area πr^2 . The cone tapers to a point, so most cross-sections are much smaller — it uses only one-third of the cylinder's space. The frustum keeps a wide base and a wide top, so it retains far more of the middle volume than the cone does, placing its capacity between the cone and the cylinder. By applying the correct formula to each shape, we can predict, compare, and design containers to hold exactly the capacity we need.
Lesson 7 Project: Model Making & Measurement — Frustum, Pyramid Frustum, and Cone Containers	Learners built three cardboard model containers (a cone, a frustum of a cone, and a frustum of a square pyramid), measured their actual dimensions, and filled them with water to determine experimental capacity — observing directly that the three shapes, even when built to similar heights and base widths, hold measurably different amounts of water, mirroring the Kericho farmers' containers.	Learners applied $V = \frac{1}{3}\pi r^2 h$ for the cone, $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ for the frustum of a cone, and $V = \frac{1}{3}h(a^2 + b^2 + ab)$ for the frustum of a pyramid to compute theoretical capacity in cm^3 and litres; they converted between cm^3 and mL ($1 \text{ cm}^3 = 1 \text{ mL}$); and they quantified the difference between theoretical and experimental capacity as a percentage discrepancy, identifying measurement and construction error as the cause.	Learners can now explain that the Kericho tea farmers' cone, bucket (frustum of cone), and cylindrical drum hold such different amounts not because of how similar they look on the outside, but because their internal three-dimensional shapes obey fundamentally different volume formulae — and that even a 1 cm error in manufacturing a container's height or radius produces a calculable and significant change in the amount of tea it can carry.
Lesson 8 Model Building Culmination, Final Explanation & Assessment: The Kericho Farmers' Containers Solved	We observed five project groups present frustum-shaped container models — some shaped as frustums of cones, others as frustums of square pyramids — each filled with water to physically measure capacity. We recorded their target capacity, calculated capacity (from the frustum formulae) and measured capacity (from water displacement), and computed the percentage error for each group.	We learned that: (1) a cone and a cylinder with the same base radius and height always satisfy $V_{\text{cylinder}} = 3 \times V_{\text{cone}}$, which is why the Kericho cylindrical drum holds exactly three times the conical sisal cover; (2) the frustum bucket's volume — $V = (1/3)\pi h(R^2 + Rr + r^2)$ — falls between those of the cone and the cylinder because it retains a wide base while tapering, accumulating more volume than the cone but less than the full cylinder; (3) the frustum formula is derived by subtracting a smaller cone from a larger cone, and the same logic applies to pyramid frustums; (4) we can reverse-engineer the formula to find an unknown dimension (such as R) when a target capacity is specified.	We can now fully explain the anchoring phenomenon: the cylindrical drum, the conical sisal cover and the metal frustum bucket look the same size because they share the same height and opening diameter, but their internal geometries — how cross-sectional area accumulates from base to top — differ fundamentally. The cylinder maintains a constant circular cross-section all the way up, so it accumulates the maximum possible volume. The cone tapers linearly to a point, cutting its volume to exactly one-third. The frustum keeps a wide bottom (radius R) and narrows only to a smaller circle (radius r, not zero), so it captures more volume than the cone but less than the cylinder. A Kericho farmer who needs a container of exactly 10 litres can use $V = (1/3)\pi h(R^2 + Rr + r^2)$, fix h and r to match available sheet-metal sizes, and solve for R — turning mathematics into a practical design tool.

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